Data Preprocessing, Evaluation of Classification & EDA

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- Data preprocessing:
  Data cleaning – data integration – Data Reduction, Data Transformation and Data Discretization.
- Evaluation of classification methods – Confusion matrix, Students T-tests and ROC curves-Exploratory
- Data Analysis – Basic tools (plots, graphs and summary statistics) of EDA, Philosophy of EDA – The Data Science Process.

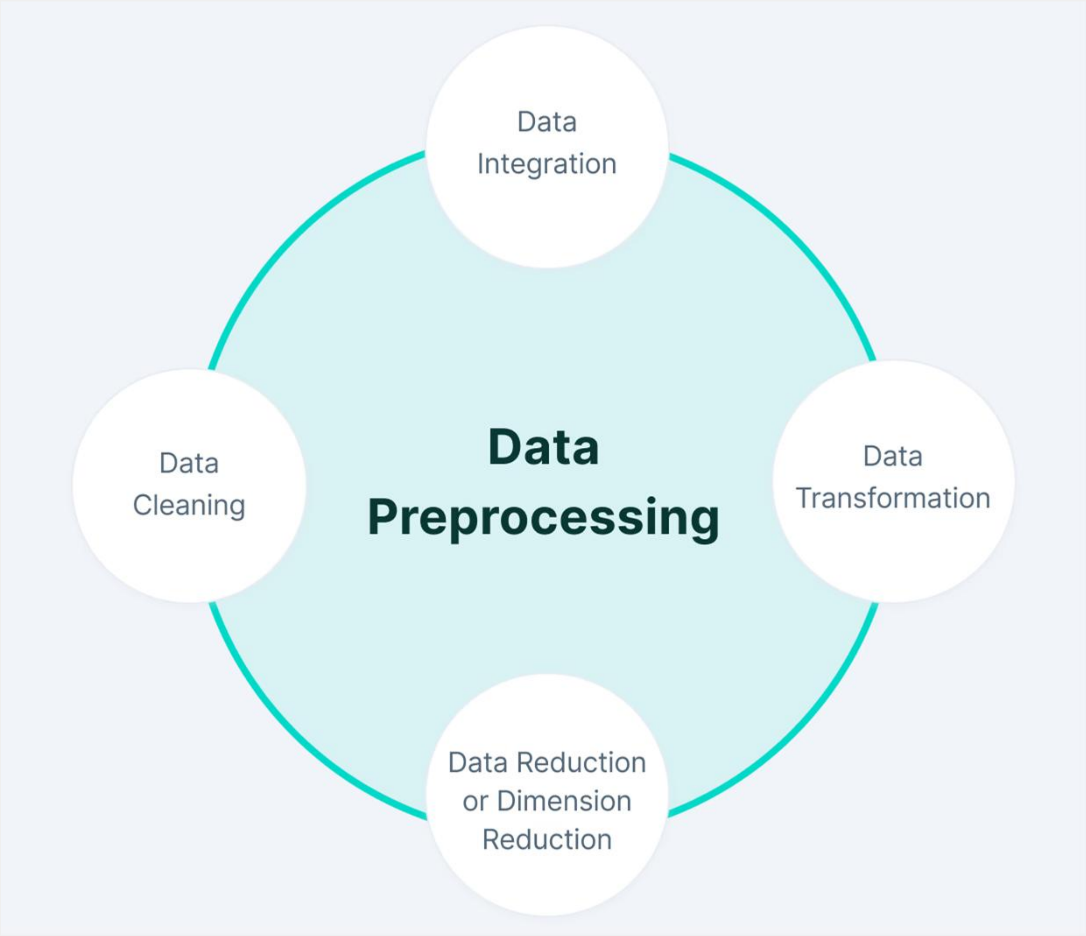
Data Preprocessing

- Real-world data is often incomplete, noisy, and inconsistent, which can lead to incorrect results if used directly. Data preprocessing in data mining is the process of cleaning and preparing raw data so it can be used effectively for analysis and model building.
  - Real data contains missing and incorrect values
  - Data may come from multiple sources
  - Large datasets often have irrelevant information
  - Clean data gives better mining results

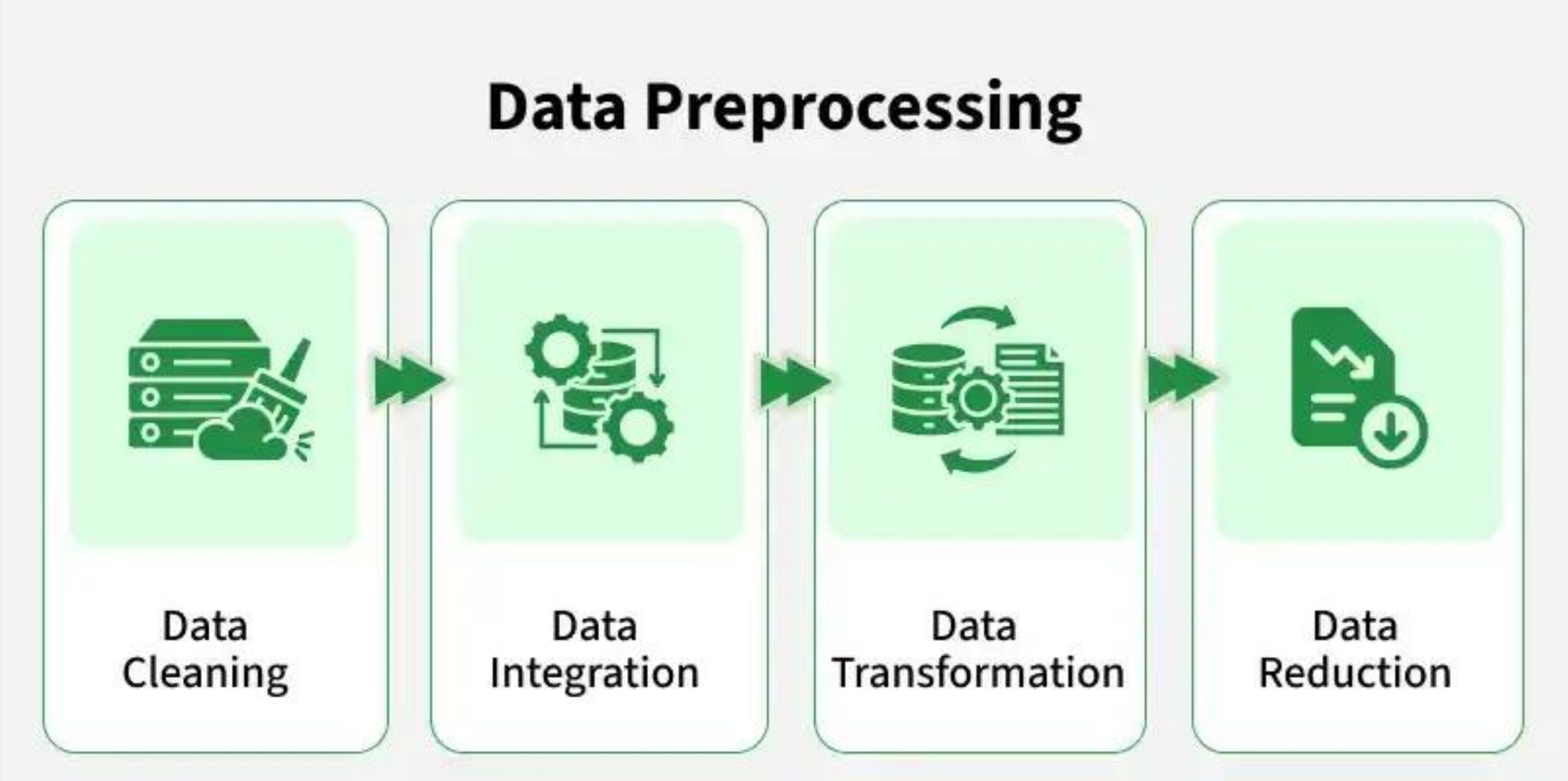
Data Preprocessing

- Data preprocessing is the crucial, preliminary step in data analysis and machine learning that transforms raw, unstructured data into a clean, organized format. It involves techniques like cleaning (handling missing values/noise), transformation (normalization/scaling), and reduction to improve data quality, accuracy, and model performance.
- Without this process, models may produce inaccurate, unreliable results due to poor-quality data. Examples include normalizing financial data, tokenizing text for Natural Language Processing (NLP), and reducing noise in images.

Data Preprocessing



Steps of Data Preprocessing



Data Cleaning

- It is the process of identifying and correcting errors or inconsistencies in the dataset. Its common tasks include:
  - Handling missing values
  - Removing duplicate records
  - Correcting wrong or inconsistent data
  - Handling Outliers

Data Cleaning

Data Cleaning Techniques

01 Remove duplicates

02 Detect & remove Outliers

03 Remove irrelevant data

04 Standardize capitalization

05 Convert data type

06 Clear formatting

07 Fix errors

08 Language translation

09 Handle missing values

Techniques used in Data Cleaning

Techniques used:

- Mean Imputation: Replaces missing values with the average of the attribute.
- Median Imputation: Replaces missing values with the middle value, useful when outliers exist.
- Mode Imputation: Replaces missing values with the most frequent value.
- Deletion Method: Removes records that contain missing values.
- Interquartile Range (IQR): Detects outliers using the range between Q1 and Q3.
- Z-Score Method: Identifies outliers based on standard deviation from the mean.
- Binning: Smooths noisy data by grouping values into bins.
- Regression Smoothing: Uses regression to predict and smooth noisy values.
- Duplicate Detection: Identifies and removes repeated records.

Example

- Replacing missing age values with the average age
- Removing repeated rows in a dataset

Sample table Example

Student_ID	Name	Age	Gender	Marks	Email
101	Riya	20	F	85	riya@gmail.com
102	Aman		M	90	aman@gmail
103	NULL	21	Male	-5	dev@gmail.com
103	Dev	21	Male	-5	dev@gmail.com
104	Shreya	200	F	88	shreya@gmail.com
105	Karan	19	m	450	karan@gmail.com
106	Pooja	20	Female	92	

Sample table Example

- Problems in Raw Data
- Missing value → Age (Student 102)
- Invalid email → "aman@gmail"
- NULL name (Student 103)
- Duplicate record → Student_ID 103
- Invalid marks → -5 and 450
- Invalid age → 200
- Inconsistent gender → F, M, Male, m, Female
- Missing email → Student 106

Sample table Example

Student_ID	Name	Age	Gender	Marks	Email
101	Riya	20	Female	85	riya@gmail.com
102	Aman	20	Male	90	aman@gmail.com
103	Dev	21	Male	0	dev@gmail.com
104	Shreya	20	Female	88	shreya@gmail.com
105	Karan	19	Male	100	karan@gmail.com
106	Pooja	20	Female	92	Not Available

Changes made

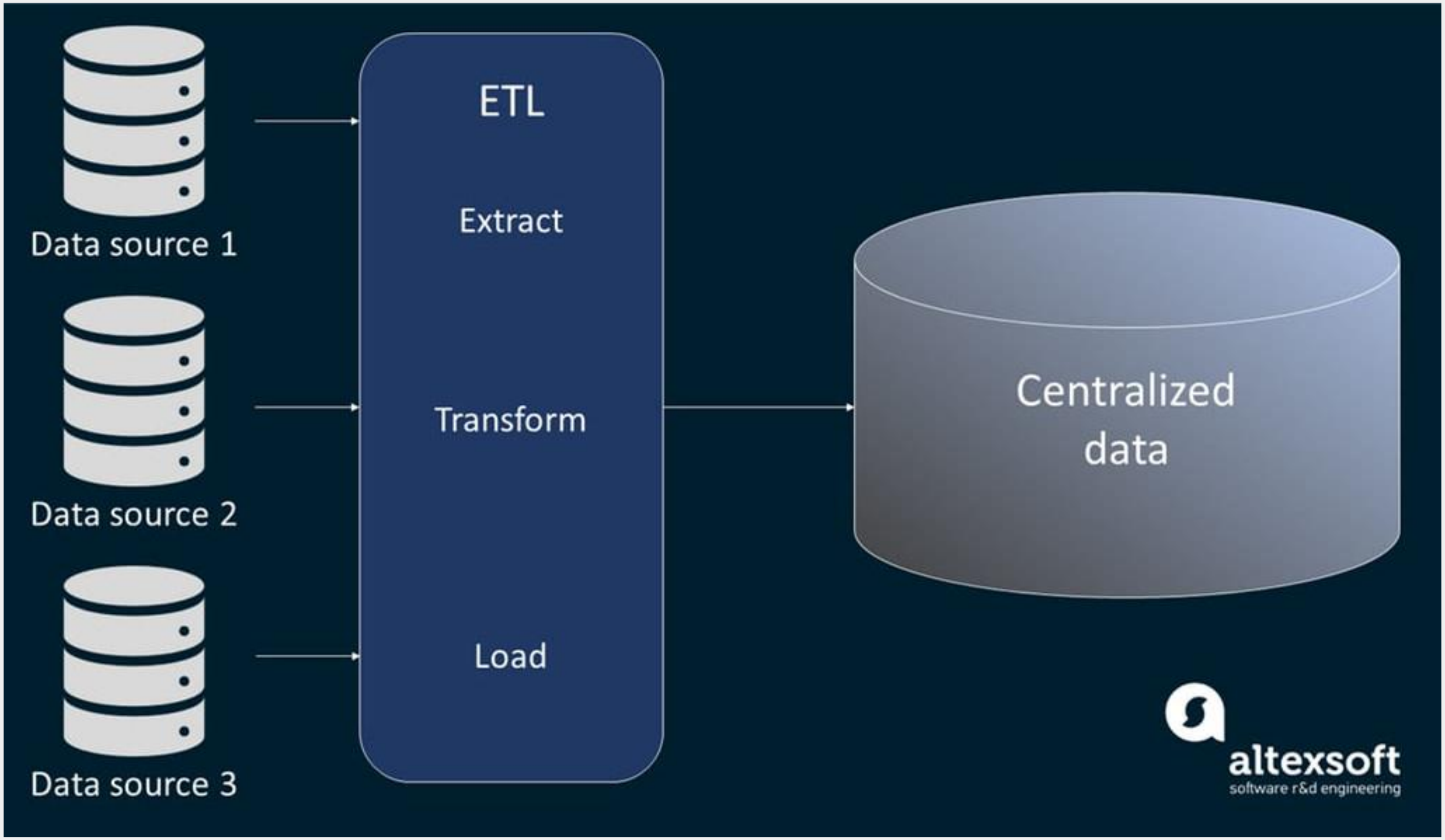
Problem	Cleaning Technique Used
Missing Age	Replaced with mean age (20)
NULL Name	Removed duplicate NULL row
Duplicate ID	Kept only valid record
Invalid Marks (-5)	Replaced with 0
Marks > 100	Capped to 100
Age = 200	Replaced with mean
Gender inconsistency	Standardized to Male/Female
Invalid email	Corrected format
Missing email	Filled with “Not Available”

Data Integration

Data Integration

- It involves merging data from various sources into a single, unified dataset. It can be challenging due to differences in data formats, structures, and meanings.
- Used when data comes from databases, files, or APIs
- Removes redundancy between datasets
- Resolves conflicts in data values

Data Integration



Techniques used for Data Integration

Techniques used:

- Schema Matching: Aligns attributes from different data sources.
- Entity Resolution: Identifies records that refer to the same real-world entity.
- Correlation Analysis: Finds and removes redundant attributes.
- Data Conflict Resolution: Resolves inconsistencies in units or data values.
- Duplicate Elimination: Removes overlapping records after integration.
- Example: Merging customer data from sales and marketing databases

Data Transformation

- Data transformation converts data into a suitable form so that data mining algorithms can work effectively.
- Bring data into a common format
- Improve mining efficiency
- Make data suitable for modeling

Data Transformation Techniques

- Min-Max Normalization: Scales data into a fixed range, usually 0 to 1.
- Z-Score Normalization: Transforms data using mean and standard deviation.
- Decimal Scaling: Normalizes data by moving the decimal point.
- Log Transformation: Reduces data skewness using logarithmic scaling.
- One-Hot Encoding: Converts categories into binary columns.
- Label Encoding: Assigns numeric labels to categorical values.
- Aggregation: Combines detailed data into summarized form.

Min-Max Normalization

$$X' = \frac{X - Xmin}{Xmax - Xmin}$$

Student	Marks
A	40
B	60
C	80
D	100

Student	Normalized Marks
A	(40-40)/(100-40) = 0
B	20/60 = 0.33
C	40/60 = 0.67
D	60/60 = 1

Here:

- Min = 40
- Max = 100

Z-Score Normalization

$$Z = \frac{X - \mu}{\sigma}$$

Student	Marks
A	40
B	60
C	80
D	100

Here:

- Mean (μ) = 70

Z-Score Normalization

$$\sigma = \sqrt{\frac{\sum(X-\mu)^2}{N}}$$

Marks (X)	X – 70	(X – 70)²
40	-30	900
60	-10	100
80	10	100
100	30	900

$$\sigma = \sqrt{500} = 22.36$$

Decimal Scaling

$$v' = \frac{v}{10^j}$$

Student	Marks
A	45
B	67
C	89

Find Maximum Value

Max = **89**

To make it less than 1, divide by **100**

So, j = 2

Decimal Scaling

$$v' = \frac{v}{10^j}$$

Student	Marks
A	45
B	67
C	89

Apply Formula

Student	Original	After Decimal Scaling
A	45	0.45
B	67	0.67
C	89	0.89

Solve this example with Decimal Scaling

Salary
25000
50000
75000

Data Reduction

- It reduces the dataset's size while maintaining key information. This can be done through feature selection which chooses the most relevant features and feature extraction which transforms the data into a lower-dimensional space while preserving important details.
  - Improves processing speed
  - Saves storage space
  - Makes analysis easier

Data Reduction Techniques

- Principal Component Analysis (PCA): Reduces dimensions by projecting data onto principal components.
- Linear Discriminant Analysis (LDA): Reduces dimensions while maximizing class separation.
- Filter Methods: Select features based on statistical measures.
- Wrapper Methods: Select features using model performance.
- Embedded Methods: Perform feature selection during model training.
- Simple Random Sampling: Selects data points randomly from the dataset.
- Stratified Sampling: Samples data proportionally from each class.

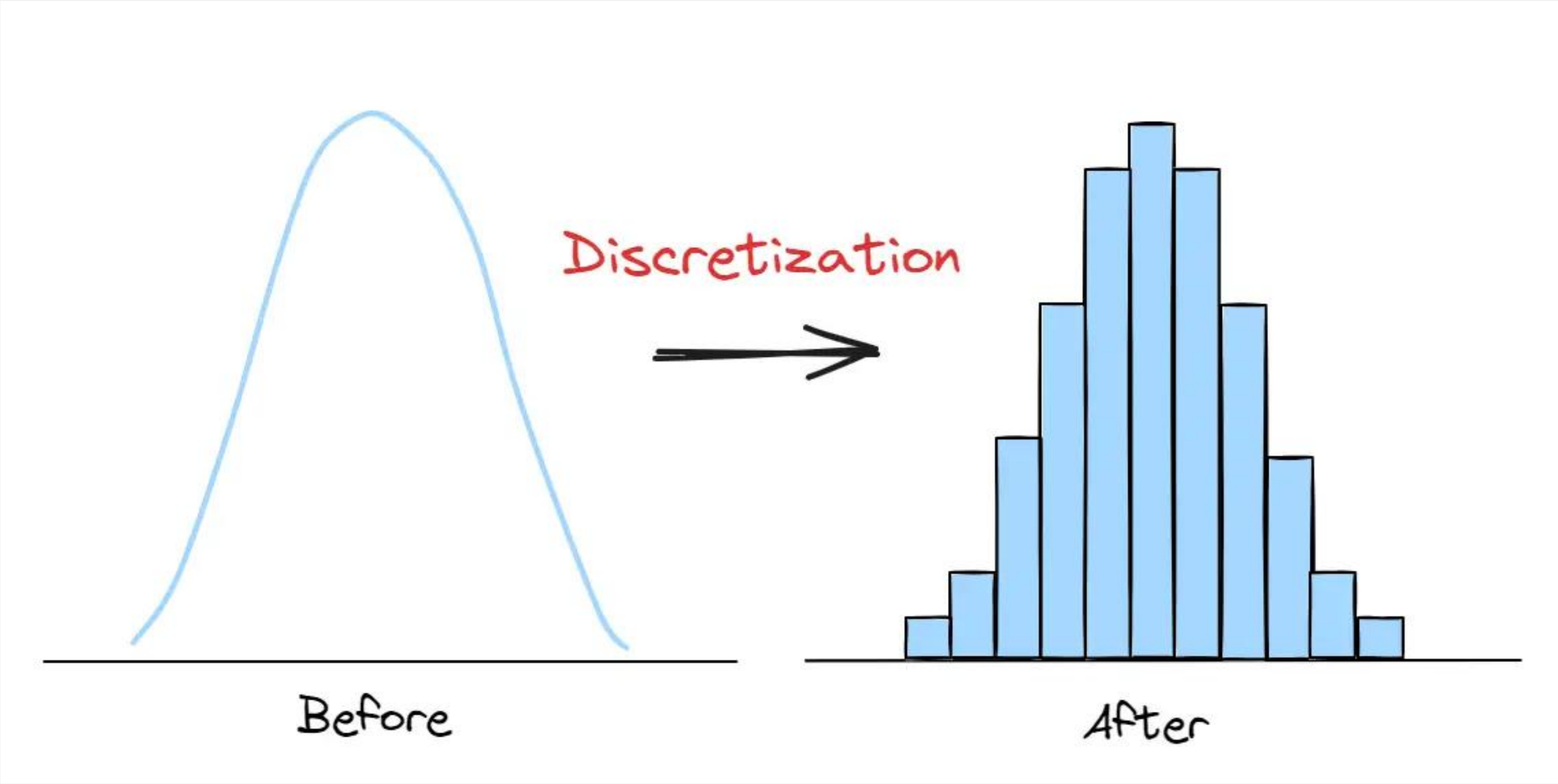
Benefits of Data Preprocessing

- Improves data quality
- Increases accuracy of mining results
- Reduces errors in models
- Makes data easier to understand

Data Discretization

- Data discretization is a preprocessing technique that converts continuous, quantitative data into discrete, categorical bins or intervals. It reduces data volume, improves model performance, and enhances interpretability by turning precise, high-cardinality numbers into meaningful, ranked groups (e.g., converting exact ages to "child," "adult," "senior"). Common techniques include binning, histogram analysis, and clustering.

Data Discretization



Data Discretization Example

Student	Age
A	18
B	22
C	35
D	42
E	60

Student	Age	Age Group
A	18	Young
B	22	Adult
C	35	Adult
D	42	Senior
E	60	Senior

We will create age groups:

- 0–20 → Young
- 21–40 → Adult
- 41–60 → Senior

Solve it!

Marks
35
55
72
90

Marks	Grade
35	?
55	?
72	?
90	?
35	?

Advantages of Data Preprocessing

- Improved Data Quality: Ensures data is clean, consistent, and reliable for analysis.
- Better Model Performance: Reduces noise and irrelevant data, leading to more accurate predictions and insights.
- Efficient Data Analysis: Streamlines data for faster and easier processing.
- Enhanced Decision-Making: Provides clear and well-organized data for better business decisions.

Advantages of Data Preprocessing

Evaluation of Classification Methods

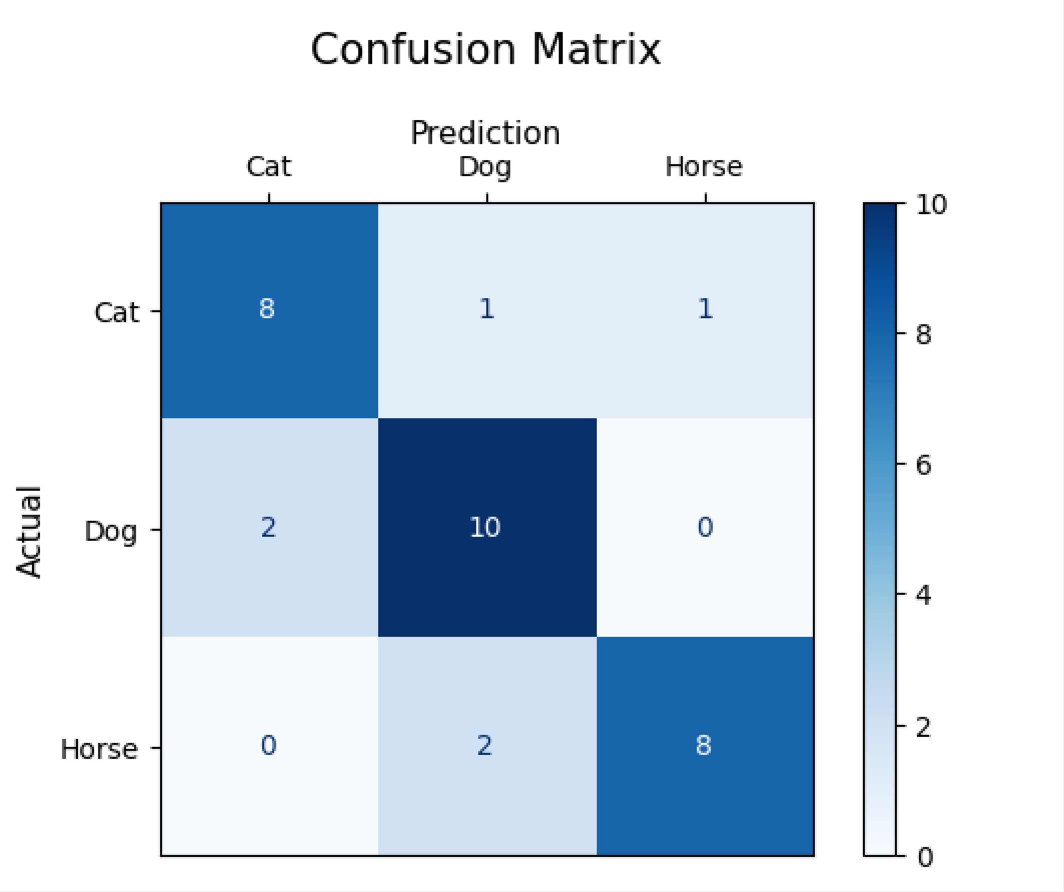
What is Evaluation of Classification Methods?

- Measuring how well a classification model is performing.

Example:

  - Spam vs Not Spam
  - Disease vs No Disease
  - Pass vs Fail

Confusion Matrix



Confusion Matrix

	Predicted Yes	Predicted No
Actual Yes	50	10
Actual No	5	35

Four Terms in Confusion Matrix

- TP (True Positive) = 50
- Correctly predicted disease
- FN (False Negative) = 10
- Disease but predicted No
- FP (False Positive) = 5
- No disease but predicted Yes
- TN (True Negative) = 35
- Correctly predicted No disease

Example: FP

- **False**

→ Model is **wrong**

- **Positive**

→ Model predicted **Positive (Yes)**

Confusion Matrix

Metric	Meaning
TP	Model predicted Positive, actually Positive
FP	Model predicted Positive, actually Negative
FN	Model predicted Negative, actually Positive
TN	Model predicted Negative, actually Negative

	Predicted Yes	Predicted No
Actual Yes	TP = 50	FN = 10
Actual No	FP = 5	TN = 35

Metrics in Confusion Matrix

$$Accuracy = \frac{TP + TN}{Total}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Student's T-Test

Student’s T-Test

- A t-test (also known as Student's t-test) is a tool for evaluating the means of one or two populations using hypothesis testing. A t-test may be used to evaluate whether a single group differs from a known value (a one-sample t-test), whether two groups differ from each other (an independent two-sample t-test), or whether there is a significant difference in paired measurements (a paired, or dependent samples t-test).

α Value	Meaning	Usage
0.10	10% error allowed	Exploratory studies
0.05	5% error allowed	Most common in research
0.01	1% error allowed	Very strict studies

Pre-defined T-Table

df	α = 0.10	α = 0.05	α = 0.02	α = 0.01
1	6.314	12.706	31.821	63.657
2	2.920	4.303	6.965	9.925
3	2.353	3.182	4.541	5.841
4	2.132	2.776	3.747	4.604
5	2.015	2.571	3.365	4.032
6	1.943	2.447	3.143	3.707
7	1.895	2.365	2.998	3.499
8	1.860	2.306	2.896	3.355
9	1.833	2.262	2.821	3.250
10	1.812	2.228	2.764	3.169
20	1.725	2.086	2.528	2.845
30	1.697	2.042	2.457	2.750
∞	1.645	1.960	2.326	2.576

Example

- A teacher claims that the average marks of students is 50.
- A sample of 5 students has marks:

52, 48, 55, 49, 46

- At $\alpha = 0.05$, test whether the true mean is different from 50.

Solution

Step 1: State hypothesis

$H_0 : \mu = 50$

$H_1 : \mu \neq 50$ (two-tailed) → difference in both directions

Step 2 finding Mean value

$$\bar{x} = \frac{52+48+55+49+46}{5}$$

$\bar{x} = 50$

Sample Mean is = 50

Step 3 Calculate t-value

Formula $t = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$

here $\bar{x} = 50$, $\mu = 50$ } thats why we didn't calculate SD.

So

$$t = \frac{50-50}{\text{Any value it comes}}$$

$t = 0$

Step 4: Find critical value

$n = 5$

Df / Degree of freedom

= total Samples − 1 = 5−1 = 4

so critical value by Df = 4 & $\alpha = 0.05$

from t table will be

≈ 2.776

Solution

Step 5: Compare

$|t| = 0$

$0 < 2.776$

Hence, it fails to reject

Teacher's claim is correct

Solve it yourself!

- A company claims that a machine fills bottles with an average of 500 ml.
- You suspect the machine may be filling too much or too little.
- You collect a sample of 6 bottles:
- 502, 498, 505, 497, 501, 496
- Test at $\alpha = 0.05$ whether the true mean is different from 500 ml.

Solve it yourself!

Step 1 hypothesis forming

$H_0 : \mu = 500$

$H_1 : \mu \neq 500$ (two-tailed)

Step 2 Calculate sample mean

$$\text{mean} / \bar{x} = \frac{502+498+505+497+501+496}{6}$$

$$= 2999/6 = 499.83$$

$$\bar{x} = 499.83$$

since $\bar{x}$ and μ are not equal so calculate SD

Step 3 Calculate SD

$$s = \sqrt{\frac{\Sigma (x-\bar{x})^2}{n-1}}$$

after calculation we get

$$s \simeq 3.43$$

Step 4: Calculate T-value

$$\text{Formula} \quad t = \frac{\bar{x}-\mu}{s/\sqrt{n}}$$

$$t = \frac{499.83-500}{3.43/\sqrt{6}}$$

$$t = \frac{-0.17}{3.43/2.45}$$

Solve it yourself!

$$t = \frac{-0.17}{1.40}$$

$$t \simeq -0.12$$

Step 5: Find critical value

$n = 6$

$df = n - 1 = 5$

$\alpha = 0.05$

so critical value will be

$\simeq 2.571$

Step 6: Decision

$|t| = 0.12$

$0.12 < 2.571$

fail to reject ✓ correct

Solve it yourself!

- A factory claims that the average lifetime of its batteries is 100 hours.
- A random sample of 8 batteries gives the following lifetimes (in hours):

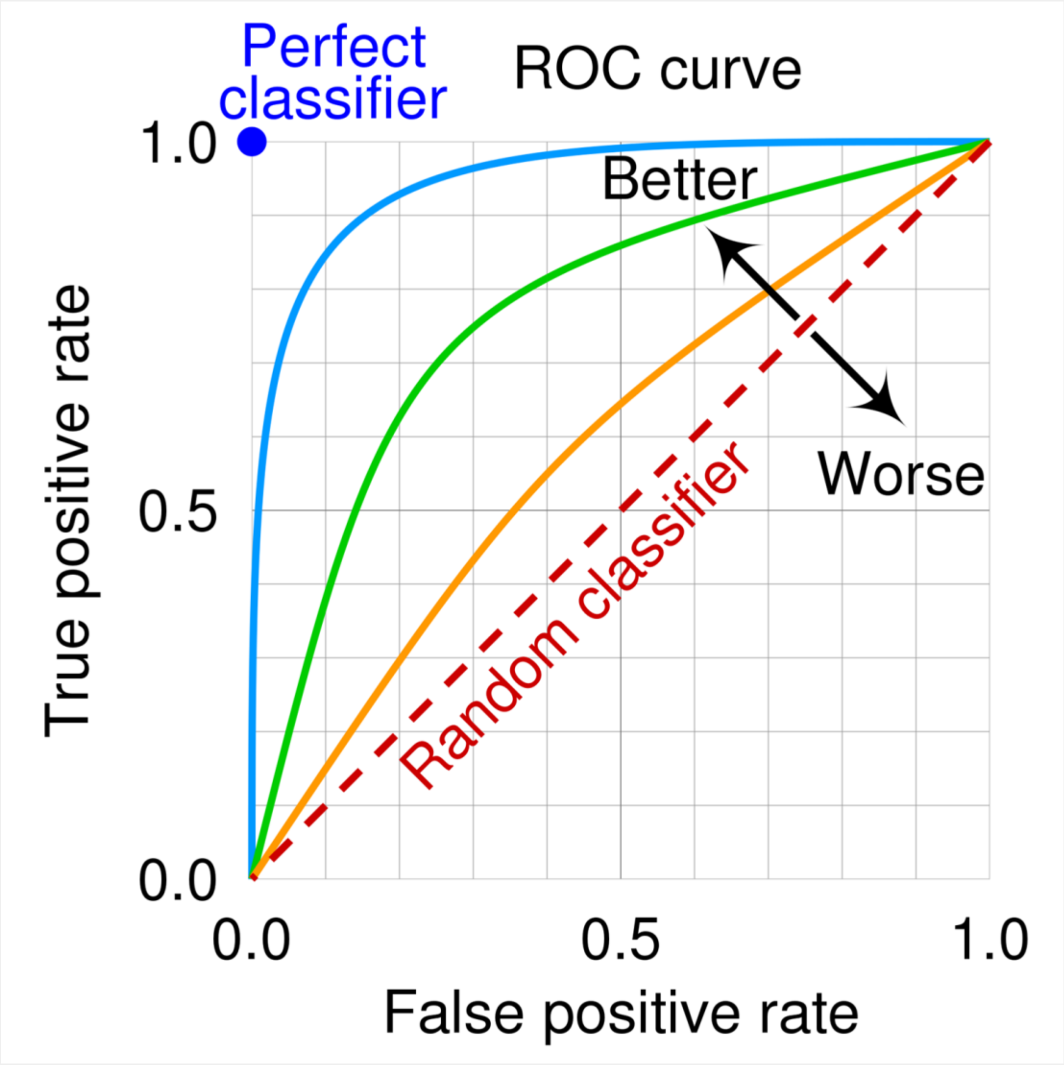
98, 102, 95, 101, 99, 97, 103, 96

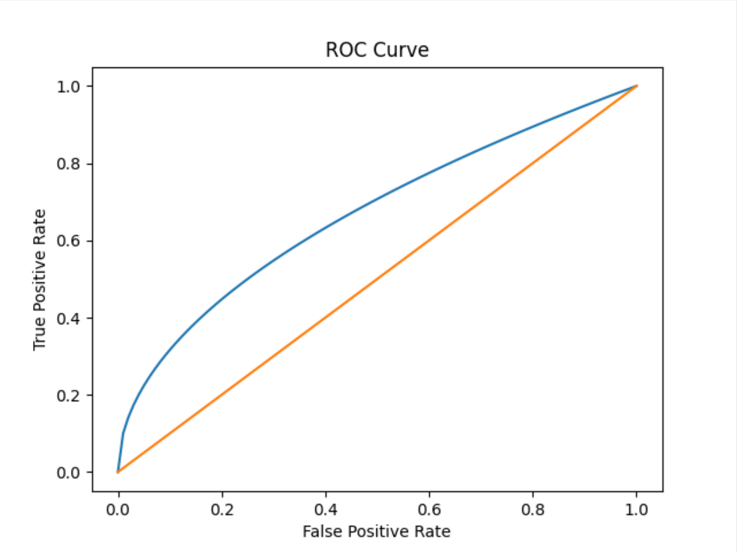
- At $\alpha = 0.05$, test whether the true mean lifetime is different from 100 hours.

ROC Curve

- The Receiver Operating Characteristic (ROC) curve is a graphical evaluation tool used to assess the performance of a binary classification model.
- It illustrates the trade-off between the True Positive Rate (TPR), also known as sensitivity or recall, and the False Positive Rate (FPR) at various threshold settings.
- The TPR measures the proportion of actual positive cases that are correctly identified by the model, while the FPR measures the proportion of actual negative cases that are incorrectly classified as positive.
- By plotting FPR on the x-axis and TPR on the y-axis, the ROC curve shows how well the classifier distinguishes between the two classes.

ROC Curve


Perfect classifier
ROC curve
Better
Worse
Random classifier
1.0
0.5
0.0
0.0
0.5
1.0
True positive rate
False positive rate


ROC Curve
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
True Positive Rate
False Positive Rate

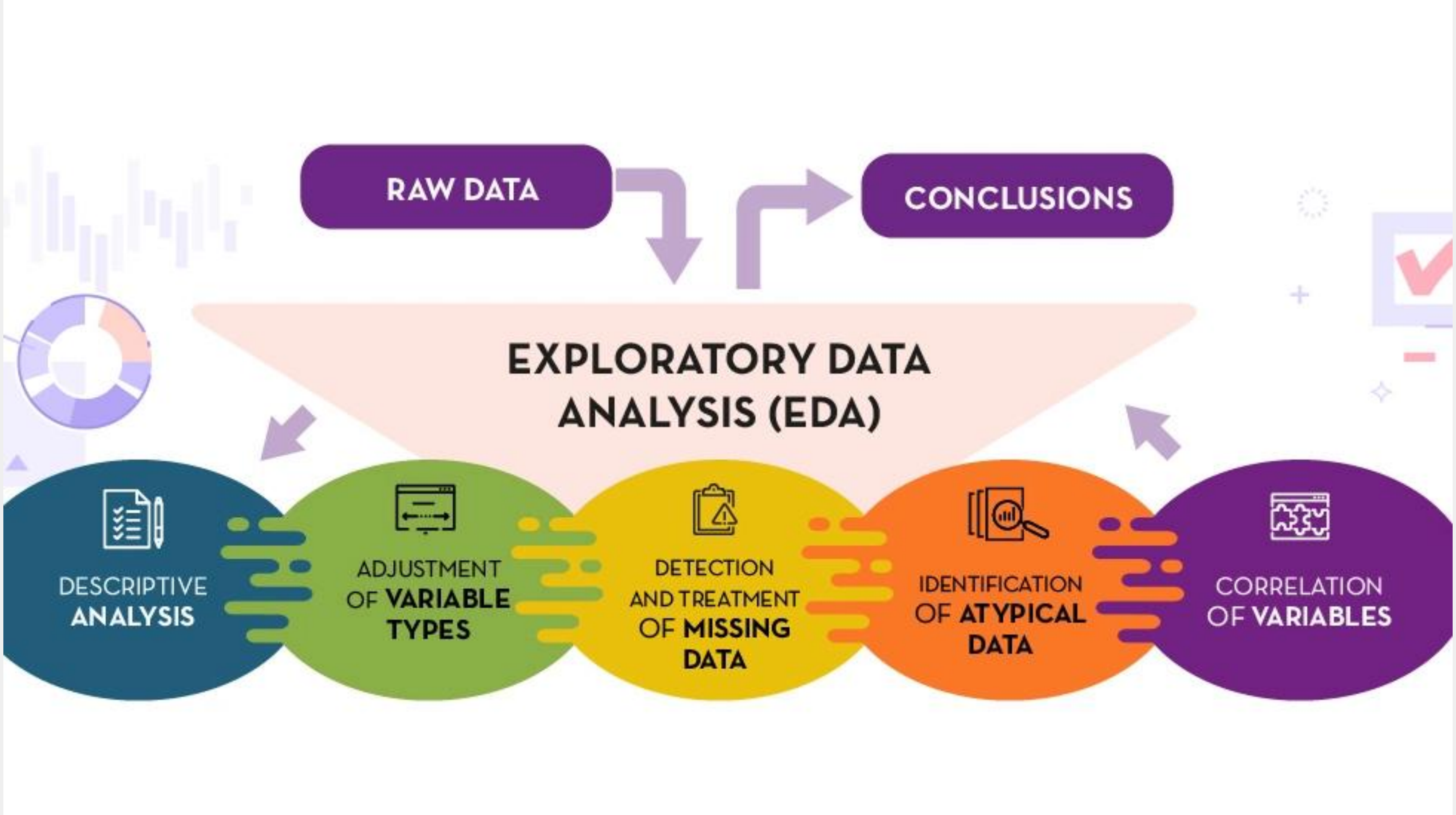
ROC Curve

- An important metric associated with the ROC curve is the Area Under the Curve (AUC). The AUC provides a single numerical value representing the model’s overall performance.
- A model with an AUC value close to 1 indicates excellent discriminative ability, whereas an AUC of 0.5 suggests that the model performs no better than random guessing.
- ROC curves are particularly useful in fields such as medical diagnosis, fraud detection, and risk prediction, where evaluating the trade-off between sensitivity and specificity is critical.

EDA (Exploratory Data Analysis)

- Exploratory Data Analysis (EDA) is the process of analyzing datasets to summarize their main characteristics, often using visual and statistical methods.
- It is performed before applying formal modeling or hypothesis testing techniques.
- The primary goal of EDA is to understand the structure of the data, identify patterns and relationships, detect anomalies or outliers, and verify assumptions.
- By exploring the dataset, researchers gain insights that guide further analysis and model development.

EDA (Exploratory Data Analysis)

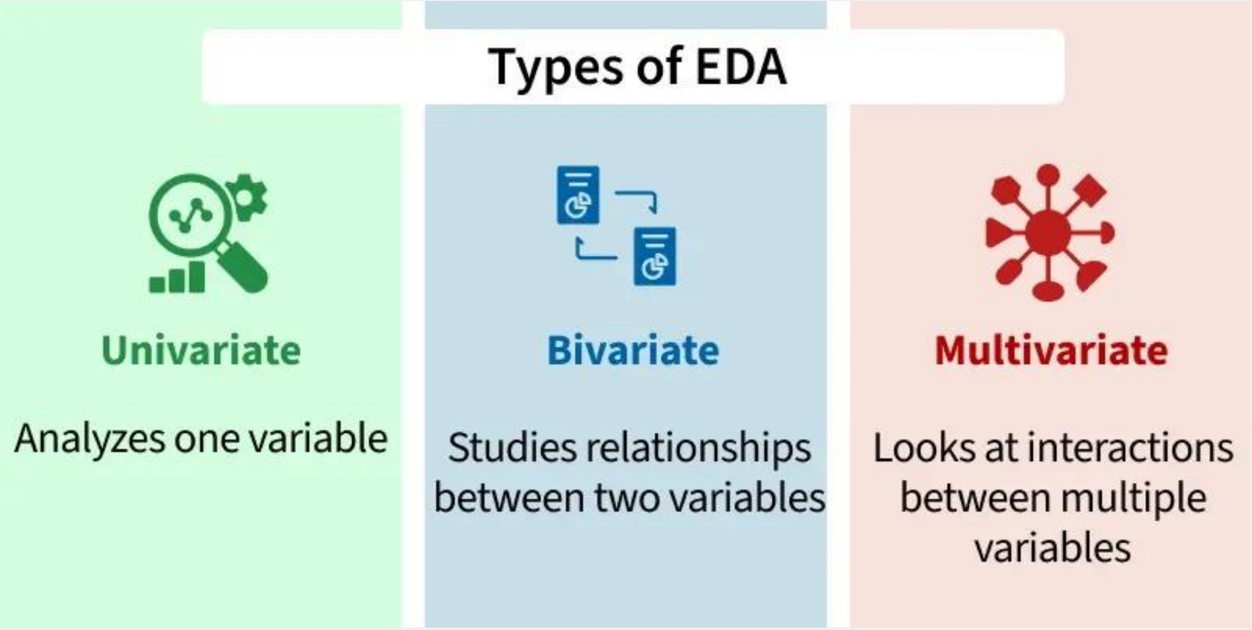


Key Aspects of EDA

- **Data Cleaning & Preprocessing:** Identifying missing values, duplicate data, and structural errors.
- **Descriptive Statistics:** Summarizing data using metrics like mean, median, mode, and standard deviation.
- **Data Visualization:** Using charts such as histograms, scatter plots, box plots, and heatmaps to visualize distributions and relationships.
- **Outlier Detection:** Identifying anomalies that could skew analysis.
- **Correlation Analysis:** Understanding relationships between variables (e.g., Pearson correlation).

Main Techniques Used in EDA

- **Univariate Analysis:** Analyzing a single variable to understand its distribution (e.g., histograms).
- **Bivariate/Multivariate Analysis:** Investigating relationships between two or more variables (e.g., scatter plots, heatmaps).



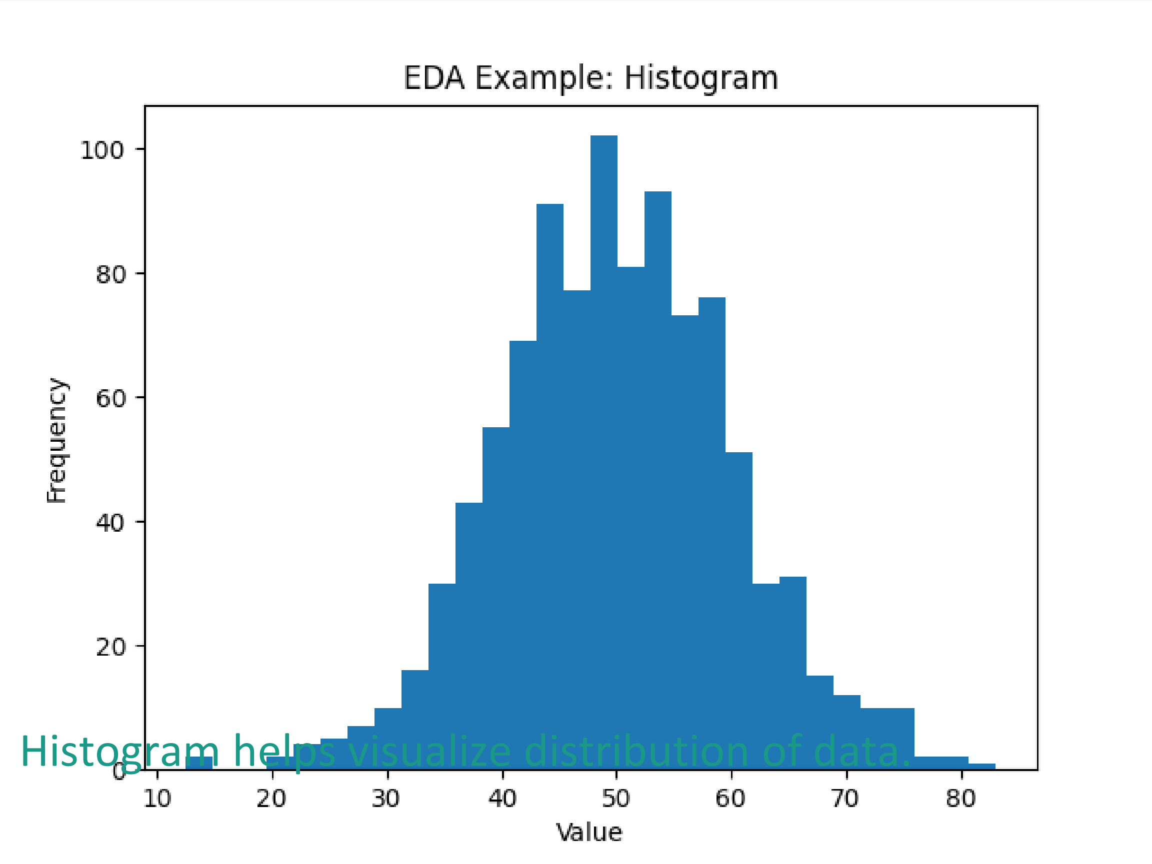
Philosophy of Exploratory Data Analysis

- The philosophy of Exploratory Data Analysis emphasizes understanding data before applying complex statistical models.
- This approach was strongly promoted by John Tukey, who believed that data analysis should focus on discovering patterns and insights rather than merely confirming predefined hypotheses.
- According to this philosophy, analysts should allow the data to reveal its structure through visualization and open-minded exploration.

Basic Tools of EDA

- Plots: Histogram, Boxplot, Scatterplot.
  - Graphs: Line charts, Bar charts.
  - Summary statistics: Mean, Median, Std Dev.

EDA Histogram Example



Histogram helps visualize distribution of data.

The Data Science Process

- The Data Science Process is a structured framework that guides the transformation of raw data into actionable insights.
- It typically begins with problem definition, where the objective and scope of the analysis are clearly identified.
- This is followed by data collection from relevant sources.
- Once the data is gathered, data cleaning and preprocessing are performed to handle missing values, inconsistencies, and noise.

The Data Science Process

- After cleaning, Exploratory Data Analysis is conducted to understand patterns, correlations, and distributions.
- Feature engineering is then applied to create meaningful variables that improve model performance.
- The next stage involves model building using suitable machine learning or statistical techniques.
- Model evaluation follows, where performance metrics such as accuracy, precision, recall, confusion matrix, and ROC curves are used to assess effectiveness.
- Finally, the model is deployed in a real-world environment and continuously monitored to ensure sustained performance.

The Data Science Process

- The Data Science Process ensures that analysis is systematic, reproducible, and aligned with real-world objectives.
- It integrates statistical reasoning, computational tools, and domain knowledge to derive meaningful insights from data.

The Data Science Process

- 1. Problem Definition
  - 2. Data Collection
  - 3. Data Preprocessing
  - 4. EDA
  - 5. Modeling
  - 6. Evaluation
  - 7. Deployment

Thank You!